

Enterprise File Sharing & FTP Server

CentOS 7 · Samba · vsftpd

Project Title	Enterprise File Sharing & FTP Server
Operating System	CentOS 7
Server IP Address	192.168.10.2
Network	192.168.10.0/24
File Sharing Service	Samba
FTP Service	vsftpd
Virtualization Platform	VMware
Author	Hassan Khan

1 Executive Summary

This project focuses on the implementation and administration of an enterprise-style file-sharing and file-transfer environment using a CentOS 7 Linux server.

The primary objective was to configure a centralized server that provides secure and authenticated file-sharing services to network clients. Two major services were implemented:

1. **Samba** for network-based file sharing and collaboration.
2. **vsftpd** for FTP-based file transfers.

The project also involved configuring Linux users and groups, file and directory permissions, firewall rules, SELinux policies, authentication, access controls, client connectivity, and troubleshooting procedures.

The completed environment successfully allowed authorized users to access shared files through Samba and transfer files through FTP while restricting unauthorized access.

This project demonstrates practical skills in Linux system administration, network services, access control, security configuration, troubleshooting, and technical documentation.

2 Introduction

File sharing and file transfer are essential services in modern IT infrastructure. Organizations often require centralized systems that allow employees and users to securely access, share, upload, and download files across a network.

In this project, a CentOS 7 server was configured as a centralized file-sharing and FTP server. The server was connected to a private network and assigned the static IP address:

```
192.168.10.2
```

Samba was used to provide Windows-compatible network file sharing, while vsftpd was used to provide FTP-based file transfer services.

The project also focused on implementing appropriate security controls to ensure that only authorized users could access shared resources.

3 Project Objectives

The main objectives of this project were:

- To prepare a CentOS 7 server for enterprise file-sharing services.

- To configure a static IP address for the server.
- To create and manage Linux users.
- To create and manage user groups.
- To configure Samba for network file sharing.
- To create a centralized Samba shared directory.
- To configure authenticated access to Samba.
- To configure an FTP server using vsftpd.
- To enable authenticated FTP access.
- To test FTP file uploads and downloads.
- To configure Linux file and directory permissions.
- To configure Samba access control.
- To configure FTP user restrictions.
- To configure the CentOS firewall.
- To configure SELinux policies for Samba and FTP.
- To test client-server connectivity.
- To troubleshoot common Samba and FTP problems.
- To document the complete implementation process.

4 Project Scope

The scope of this project included the configuration and testing of a Linux-based enterprise file-sharing environment.

The project covered:

- CentOS 7 server preparation.
- Network configuration.
- User and group management.
- Samba installation and configuration.
- FTP installation and configuration.
- Client configuration.
- File access testing.
- Security and permissions.
- Firewall configuration.
- SELinux configuration.
- Troubleshooting.
- Documentation.

The project was implemented in a virtualized lab environment using VMware.

5 Lab Environment

5.1 Hardware and Virtualization

The project was implemented using a virtual machine environment.

Component	Configuration
Virtualization Platform	VMware
Server OS	CentOS 7
Client OS	Windows/Linux Client
Network Type	Virtual Network
Server IP	192.168.10.2
Network	192.168.10.0/24

The CentOS server was configured with two network interfaces.

The primary lab interface was:

```
ens33
IP Address: 192.168.10.2/24
```

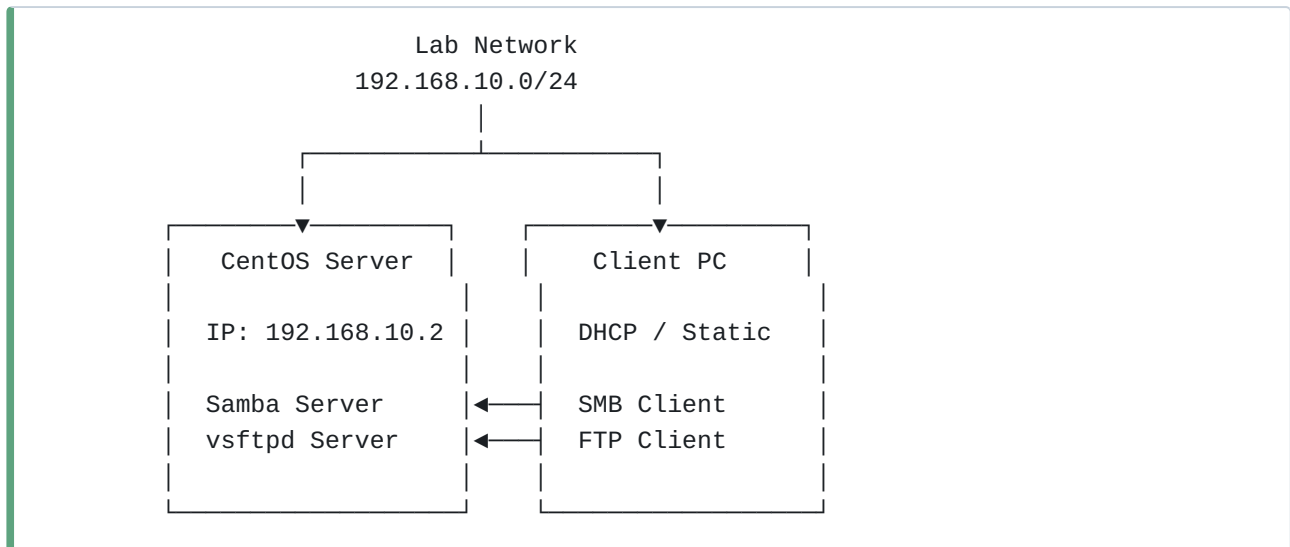
A secondary interface was available for external connectivity:

```
ens36
IP Address: 192.168.149.130/24
```

The main project services operated on the 192.168.10.0/24 lab network.

6 Network Topology

The project followed a simple client-server architecture.



The CentOS server provided:

- Samba file sharing.
- FTP file transfer.
- User authentication.
- Access control.
- Centralized file storage.

The client was used to test:

- Samba file access.
- File creation.
- File reading.
- FTP login.
- FTP uploads.
- FTP downloads.

7 CentOS Server Preparation

The CentOS 7 server was prepared as the central file-sharing server.

The server network configuration was verified using:

```
ip addr
```

The primary network interface was identified as:

```
ens33
```

The server was configured with:

```
IP Address: 192.168.10.2
Subnet Mask: 255.255.255.0
Network: 192.168.10.0/24
```

Network connectivity was tested to ensure that the server could communicate with client systems.

8 User and Group Management

A dedicated group named `fileshare` was created to simplify access management.

```
fileshare
```

The following users were configured:

```
hassan
ali
```

The users were added to the `fileshare` group.

The group-based access structure was:

```
fileshare
|
├─ hassan
└─ ali
```

Group membership was verified using:

```
getent group fileshare
```

User membership was verified using:

```
id hassan
```

```
id ali
```

This approach allowed access permissions to be managed centrally through group membership.

9 Samba File Sharing

Samba was configured to provide network file sharing.

The shared directory was:

```
/srv/samba/shared
```

The Samba share was named:

```
SharedFiles
```

Clients could access the share using:

```
\\192.168.10.2\SharedFiles
```

The Samba configuration included access restrictions based on the `fileshare` group.

Example configuration:

```
[SharedFiles]
    path = /srv/samba/shared
    browseable = yes
    writable = yes
    read only = no
    valid users = @fileshare
    force group = fileshare
    create mask = 0660
    directory mask = 2770
```

The configuration was validated using:

```
testparm
```

The Samba service was managed using:

```
systemctl start smb
systemctl enable smb
systemctl restart smb
```

The service status was verified using:

```
systemctl status smb
```

10 Samba Permissions and Access Control

The Samba shared directory was configured with restricted permissions.

The directory ownership was:

```
root:fileshare
```

The directory permissions were configured as:

```
2770
```

The permission structure ensured that:

- The owner had full access.
- Members of the `fileshare` group had full access.
- Other users had no access.

The permissions were verified using:

```
ls -ld /srv/samba/shared
```

The access control was further strengthened using:

```
valid users = @fileshare
```

This ensured that only authorized members of the `fileshare` group could access the shared directory.

11 FTP Server Configuration

The vsftpd service was configured to provide FTP-based file transfers.

The FTP server was accessible through:

```
192.168.10.2
```

The standard FTP port was:

```
21
```

The following configuration options were used:

```
anonymous_enable=NO
local_enable=YES
write_enable=YES
chroot_local_user=YES
allow_writeable_chroot=YES
```

The configuration provided the following security features:

- Anonymous FTP access disabled.
- Local user authentication enabled.
- File uploads enabled.

- File downloads enabled.
- Users restricted to their FTP environment.

The FTP service was managed using:

```
systemctl start vsftpd  
systemctl enable vsftpd  
systemctl restart vsftpd
```

The service status was verified using:

```
systemctl status vsftpd
```

Port 21 was verified using:

```
ss -tulpn | grep :21
```

12 Client Configuration

The client system was configured to communicate with the CentOS server over the `192.168.10.0/24` network.

Network connectivity was tested using:

```
ping 192.168.10.2
```

After successful network connectivity, the client was used to access the Samba share:

```
\\192.168.10.2\SharedFiles
```

The FTP server was accessed using:

```
ftp://192.168.10.2
```

Authenticated users were used to test both services.

13 File Access Testing

The configured services were tested to verify that users could successfully access and transfer files.

13.1 Samba Read Test

A test file was created on the server:

```
samba-read-test.txt
```

The file was accessed from the client through the Samba share.

The authorized user successfully opened and read the file. — **Status: Successful**

13.2 Samba Write Test

A test file was created from the client:

```
samba-write-test.txt
```

The server was checked using:

```
ls -l /srv/samba/shared/
```

The file was successfully created on the server.

The authorized user successfully created a file through Samba. — **Status: Successful**

13.3 FTP Upload Test

A test file was uploaded from the client:

```
ftp-upload-test.txt
```

The server was checked using:

```
ls -l /home/hassan
```

The uploaded file was successfully received by the FTP server.

FTP file upload was successful. — **Status: Successful**

13.4 FTP Download Test

A test file was created on the FTP server:

```
ftp-download-test.txt
```

The client successfully downloaded the file.

FTP file download was successful. — **Status: Successful**

13.5 Unauthorized Access Test

An unauthorized user attempted to access the Samba shared directory.

The access attempt was denied because the user was not a member of the authorized `fileshare` group.

Unauthorized access was successfully restricted. — **Status: Successful**

14 Security Configuration

Security was implemented through multiple layers.

14.1 Linux File Permissions

The Samba directory used:

```
2770
```

This prevented unauthorized users from accessing the directory.

14.2 Group-Based Access

Access was controlled through:

```
fileshare
```

Only authorized users were members of this group.

14.3 Samba Access Control

The Samba configuration restricted access using:

```
valid users = @fileshare
```

14.4 FTP Security

Anonymous FTP access was disabled:

```
anonymous_enable=NO
```

Only authenticated users were permitted to access the FTP service.

14.5 Firewall

The CentOS firewall was configured to allow:

```
ftp  
samba
```

The firewall was verified using:

```
firewall-cmd --list-services
```

14.6 SELinux

SELinux policies were considered and configured for the required services.

For Samba, the shared directory used the appropriate context:

```
samba_share_t
```

FTP home-directory access was verified using:

```
getsebool ftp_home_dir
```

15 Troubleshooting

Several common problems were considered during the implementation.

Samba Troubleshooting

Potential problems included:

- Samba service not running.
- Invalid `smb.conf` configuration.
- Incorrect user credentials.
- Missing group membership.
- Incorrect file permissions.
- Firewall restrictions.
- SELinux restrictions.

The main troubleshooting commands included:

```
systemctl status smb
testparm
pdbedit -L
getent group fileshare
ls -ld /srv/samba/shared
firewall-cmd --list-services
getenforce
```

FTP Troubleshooting

Potential problems included:

- vsftpd service not running.
- FTP port 21 unavailable.
- Authentication failure.
- Upload permission errors.
- Firewall restrictions.
- SELinux restrictions.

The main troubleshooting commands included:

```
systemctl status vsftpd
ss -tulpn | grep :21
firewall-cmd --list-services
getsebool ftp_home_dir
journalctl -u vsftpd
```

Detailed troubleshooting documentation was created separately in the project folder.

16 Testing Results

The final environment was tested successfully.

Test	Result
Server Network Connectivity	Successful
Samba Service	Successful
Samba Authentication	Successful
Samba Read Access	Successful
Samba Write Access	Successful
FTP Service	Successful

FTP Authentication	Successful
FTP Upload	Successful
FTP Download	Successful
User Management	Successful
Group Management	Successful
Linux Permissions	Successful
Samba Access Control	Successful
FTP Restrictions	Successful
Firewall Configuration	Successful
SELinux Configuration	Successful
Unauthorized Access Restriction	Successful

17 Challenges and Solutions

Challenge 1: Samba Access Control

Problem

Users needed controlled access to the shared directory.

Solution

A dedicated `fileshare` group was created, and Samba access was restricted using:

```
valid users = @fileshare
```

Result

Only authorized group members could access the Samba share.

Challenge 2: File Permissions

Problem

Users required read and write access without allowing unauthorized users to access the directory.

Solution

The shared directory was configured with:

```
root:fileshare
```

and:

```
2770
```

Result

Authorized users received appropriate access while other users were denied.

Challenge 3: FTP Security

Problem

The FTP service needed to prevent anonymous access.

Solution

Anonymous FTP was disabled:

```
anonymous_enable=NO
```

Authenticated local users were enabled.

Result

Only authenticated users could access the FTP service.

Challenge 4: Firewall Configuration

Problem

Network services require firewall access to communicate with clients.

Solution

The required services were allowed through `firewalld`:

```
ftp  
samba
```

Result

Clients successfully connected to both services.

Challenge 5: SELinux

Problem

SELinux can restrict service access to directories and files.

Solution

Appropriate SELinux contexts and policies were configured for Samba and FTP.

Result

The services operated successfully while maintaining SELinux security controls.

18 Project Outcomes

The project successfully achieved its intended objectives.

The completed environment provided:

- Centralized file sharing.
- Authenticated Samba access.
- FTP file transfer.
- User and group-based access control.
- Linux permission management.
- Firewall protection.
- SELinux security.
- Client-server file access.
- File upload and download functionality.
- Unauthorized access restrictions.
- Troubleshooting documentation.

The project demonstrated the ability to deploy and manage common enterprise Linux network services in a virtualized laboratory environment.

19 Skills Demonstrated

Through this project, the following technical skills were demonstrated:

Linux Administration

- CentOS 7 administration.
- Linux command-line usage.
- Service management.

- User management.
- Group management.
- File permissions.
- Directory ownership.

Network Administration

- IP addressing.
- Network interface configuration.
- Client-server connectivity.
- Port verification.
- Firewall configuration.

File Services

- Samba configuration.
- SMB/CIFS file sharing.
- vsftpd configuration.
- FTP file transfer.

Security

- User authentication.
- Group-based authorization.
- Linux permissions.
- Samba access control.
- FTP restrictions.
- Firewall management.
- SELinux configuration.

Troubleshooting

- Service troubleshooting.
- Network troubleshooting.
- Authentication troubleshooting.
- Permission troubleshooting.
- Firewall troubleshooting.
- SELinux troubleshooting.

Documentation

- Network topology documentation.
- Configuration documentation.

- Screenshot evidence.
- Troubleshooting reports.
- Final project reporting.

20 Future Improvements

The project can be further enhanced by implementing additional enterprise-level features.

Possible improvements include:

1. Implementing secure FTP using FTPS.
2. Using SFTP instead of traditional FTP.
3. Integrating Samba with Active Directory.
4. Implementing LDAP authentication.
5. Configuring automated backups.
6. Implementing storage quotas.
7. Adding RAID-based storage.
8. Implementing centralized logging.
9. Integrating the server with a monitoring platform.
10. Implementing high availability and redundancy.

These improvements would make the environment more closely resemble a production enterprise infrastructure.

21 Conclusion

The Enterprise File Sharing and FTP Server project was successfully completed using CentOS 7.

The project demonstrated the implementation of two important network services: Samba for network file sharing and vsftpd for FTP-based file transfer.

The CentOS server was configured with a static IP address and prepared to provide centralized file services to network clients. Linux users and groups were created to manage access, while file permissions and Samba access controls were implemented to protect shared resources.

The FTP server was configured to support authenticated file transfers while disabling anonymous access. Firewall rules and SELinux policies were also configured to provide additional layers of security.

Testing confirmed that authorized users could successfully read and write files through Samba and upload and download files through FTP. Unauthorized access was restricted successfully.

Overall, this project provided practical hands-on experience in **Linux system administration, network service configuration, file sharing, FTP administration, security, access control, firewall management, SELinux, troubleshooting, and technical documentation.**

The completed project represents a practical addition to a System and Network Administration portfolio and demonstrates the ability to configure and manage essential Linux-based enterprise services.

22 Project Evidence

The project is supported by screenshots and documentation covering:

- CentOS server preparation.
- Network configuration.
- User and group management.
- Samba configuration.
- FTP configuration.
- Client configuration.
- File access testing.
- Security and permissions.
- Firewall verification.
- SELinux configuration.
- Troubleshooting procedures.

All project evidence is organized according to the project folder structure.

23 Final Project Structure

```
05-Enterprise-File-Sharing-FTP-Server
├── 01-Project-Documentation
├── 02-Network-Topology
├── 03-CentOS-Server-Preparation
├── 04-User-and-Group-Management
├── 05-Samba-File-Sharing
├── 06-FTP-Server
├── 07-Client-Configuration
├── 08-File-Access-Testing
├── 09-Security-and-Permissions
├── 10-Troubleshooting
│   ├── 01-Problem-Description.md
│   ├── 02-Samba-Troubleshooting.md
│   ├── 03-FTP-Troubleshooting.md
│   └── 04-Solutions.md
├── 11-Screenshots
├── 12-Final-Report
│   ├── Enterprise-File-Sharing-FTP-Server.docx
│   └── Enterprise-File-Sharing-FTP-Server.pdf
└── README.md
```

24 Final Status

Item	Status
Project Status	Completed
Samba File Sharing	Successfully Implemented
FTP Server	Successfully Implemented
Security Configuration	Successfully Implemented

Testing	Successfully Completed
Troubleshooting Documentation	Completed
Final Documentation	Completed
Portfolio Readiness	Ready for GitHub and CV Presentation